Introduction to Computer Vision

Spring 2026

Laboratory Work

Object Detection Overview

Student Name: Sekenova Balym
Student ID: 23B031433
Instructor: Koishiyeva Dina

KBTU

Date: March 1, 2026

Contents

1 Objective **2**

2 Task 1: Image and Bounding Box Concept **2**

3 Task 2: Object Detection Using a Pretrained Detector **3**

4 Task 3: Confidence Threshold Analysis **4**

5 Task 4: Intersection over Union (IoU) **4**

6 Task 5: Detection Result Summary **5**

7 Conclusion **5**

1 Objective

The objective of this laboratory work is to study the basic principles of object detection, understand how detection differs from image classification, and become familiar with bounding boxes, confidence scores, and detection outputs.

2 Task 1: Image and Bounding Box Concept

An RGB image containing a clearly visible dog was selected. The image was loaded and displayed. A manual bounding box was defined around the main object using the coordinate format:

$$(x, y, width, height)$$

The manually defined bounding box coordinates were:

$$(200, 120, 1150, 780)$$

The bounding box was drawn on the image to visualize object localization.

Figure 1: Original RGB image

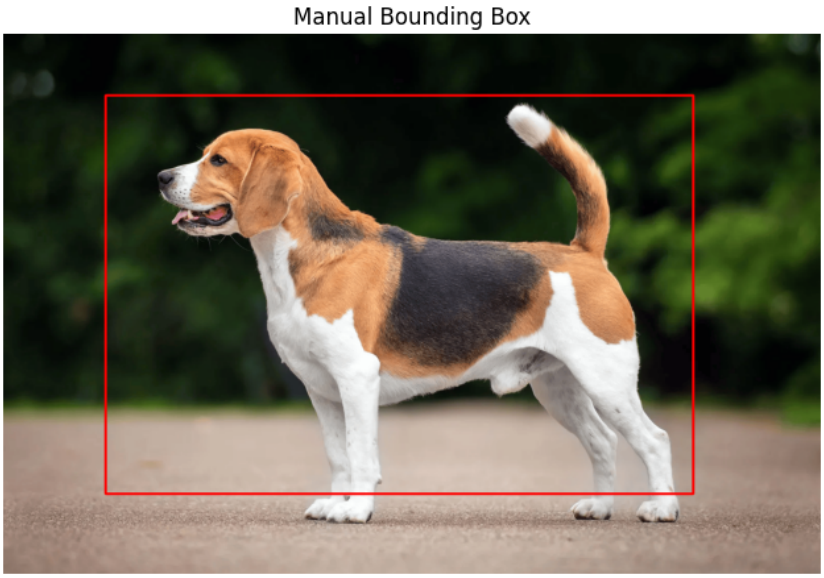

Figure 2: Manually defined bounding box

This step illustrates the concept of object localization, which differentiates object detection from simple classification.

3 Task 2: Object Detection Using a Pretrained Detector

A pretrained Faster R-CNN model with a ResNet-50 backbone was loaded from the torchvision library. The model was applied to the image. The model was pretrained on the COCO dataset, which includes a wide variety of object categories including dogs.

The detector identified the object as:

Class: dog

Confidence score: 0.999

The number of detected objects was:

1

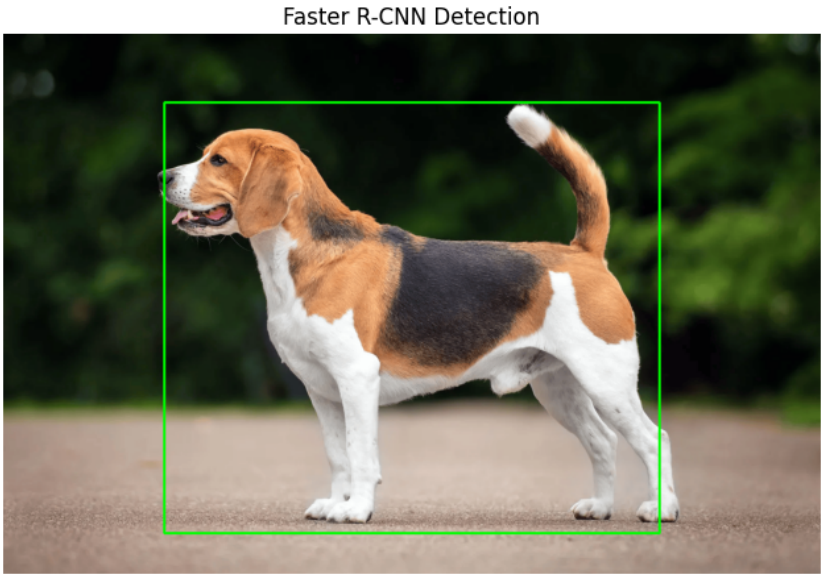

Figure 3: Detected bounding box with class label

The very high confidence score indicates that the model is highly certain about the detection.

4 Task 3: Confidence Threshold Analysis

The detector was evaluated using three different confidence thresholds:

Threshold	Detected Objects
0.25	1
0.50	1
0.75	1

The detection remained stable across all thresholds. Even at a high threshold of 0.75, the object was still detected.

This demonstrates that the detection is robust and not sensitive to confidence filtering in this case.

Lower thresholds generally increase false positives, while higher thresholds may cause missed detections. However, in this example, the object was clearly distinguishable and consistently detected.

5 Task 4: Intersection over Union (IoU)

The Intersection over Union (IoU) metric was computed between the manually defined bounding box and the detected bounding box.

$$IoU = \frac{Area\ of\ Intersection}{Area\ of\ Union}$$

The computed IoU value was:

$$IoU = 0.763$$

An IoU above 0.7 indicates good localization accuracy. The result shows strong overlap between the manually defined box and the predicted bounding box.

In object detection benchmarks, IoU ¿ 0.5 is typically considered a correct detection.

The value is not equal to 1 because manual annotation is approximate, while the detector predicts a tighter bounding box around the object boundaries.

6 Task 5: Detection Result Summary

The pretrained Faster R-CNN model successfully detected the dog in the image with very high confidence (0.999). The detection remained consistent across multiple confidence thresholds, indicating stable model performance.

The IoU value of 0.763 demonstrates accurate object localization with substantial overlap between manual and predicted bounding boxes.

Object detection quality depends on several factors:

- Object size in the image
- Background complexity
- Lighting conditions
- Object visibility and occlusion
- Model training dataset

In this case, the object was large, clearly visible, and well separated from the background, which contributed to the high detection accuracy.

7 Conclusion

This laboratory work demonstrated the difference between image classification and object detection. While classification assigns a label to an image, object detection provides both the class label and spatial localization through bounding boxes.

The pretrained detection model performed accurately, achieving high confidence and strong overlap with the manually defined bounding box. The experiment confirms that modern object detection models can reliably localize objects when image conditions are favorable.